Policy-Adherent Refund and Claims Agent Under Adversarial Persuasion

Semester-Wide Project

Course: Artificial Intelligence **Group 7**

Background and Motivation

Customer-service agents are increasingly expected to handle operational tasks such as issuing refunds, cancelling reservations, modifying orders, or processing insurance claims. These tasks appear simple, but every decision is governed by policies involving eligibility periods, required evidence, product conditions, coverage limits, previous claims, approval hierarchies, and exceptional circumstances. An incorrect decision may create financial loss, unfairly reject a genuine customer, or allow an ineligible request to bypass organisational rules.

The difficulty becomes greater because customers do not always present information in a neutral and complete manner. A genuine customer may be distressed, confused, or unable to provide all documents immediately. At the same time, a malicious user may fabricate evidence, repeatedly change the story, exploit ambiguous policy wording, or apply emotional pressure to persuade the agent to make an unauthorised exception. For example, a customer may request a refund after the permitted return period, claim that an earlier service agent had already approved it, and threaten to post negative reviews unless the refund is processed immediately. A naïve conversational agent may agree because it is trained to be helpful and accommodating, even though the requested action violates the stated policy.

A reliable agent must therefore balance several competing objectives. It should communicate politely and empathetically, identify what the customer is requesting, retrieve the relevant policy, verify the available evidence, and provide legitimate alternatives whenever possible. However, empathy should not be interpreted as automatic agreement. The system must distinguish between understanding a customer's situation and authorising an action that is not permitted. It should also recognise when a case is genuinely exceptional and requires escalation to an authorised human rather than either blindly approving or automatically rejecting the request.

In this project, you have to develop a **Policy-Adherent Refund and Claims Agent Under Adversarial Persuasion**. The system should process fictional refund, cancellation, or insurance-claim requests while interacting with both cooperative and adversarial users. It must determine eligibility using explicit policies, assess the reliability of submitted evidence, identify possible manipulation, perform the required verification steps, and decide whether to approve, reject, investigate, or escalate the request. Every decision should be accompanied by a transparent explanation grounded in the applicable policy and available evidence.

The project is primarily motivated by *CRAFT* which uses policy-aware adversarial agents that attempt to persuade customer-service agents to violate refund and cancellation rules. The work also introduces τ-break, an adversarial extension of τ-bench, and shows that common defensive strategies provide only limited protection against manipulative users (Nakash et al. 2025). This makes policy adherence under conversational pressure an important and realistic research problem. The broader customer-service setting is motivated by τ-bench, which evaluates tool-using agents in retail and airline domains where they must interact with users while following detailed operational policies. Its findings demonstrate that even strong agents may complete tasks inconsistently or perform actions that do not match the required policy-compliant outcome (Yao et al. 2024). The project is also related to the work (Fang,

Zhu, and Gurevych 2025), which investigates how potentially misaligned agent actions can be detected before execution and corrected through appropriate intervention. Together, these works motivate an agent that does not merely generate convincing responses, but verifies whether its proposed actions are authorised, evidence-supported, and safe to execute.

Module 1: Search and Constraint Satisfaction

You have to represent organisational policies, customer requests, submitted evidence, previous interactions, and decision dependencies using policy and evidence graphs. Search algorithms should be used to identify the policy clauses, documents, transaction records, and previous decisions relevant to a request. You should model the interaction between a manipulative user and a policy-defending service agent using minimax search, where the user attempts to obtain an unauthorised outcome and the agent selects responses or verification actions that maintain policy compliance. Alpha–beta pruning should be used to reduce the number of interaction strategies that must be examined. You also have to formulate request eligibility as a Constraint Satisfaction Problem by defining variables, domains, and constraints related to purchase date, cancellation period, product condition, claim coverage, submitted documents, refund limit, previous requests, and required approval. The system should identify which constraints are satisfied, violated, uncertain, or potentially manipulated.

Module 2: Planning

You have to model claim resolution as a sequential planning problem. Possible actions may include retrieving the transaction record, checking the applicable policy, requesting a receipt, validating submitted evidence, inspecting previous claims, contacting an external verification source, asking a clarification question, requesting managerial approval, issuing a refund, rejecting the request, or offering an alternative resolution. Each action should have clearly defined preconditions and effects. The planner should generate an efficient sequence of verification and resolution steps while ensuring that irreversible actions are performed only after all necessary conditions have been satisfied. It should also revise the plan when new evidence is submitted, the customer's account changes, a verification step fails, or the case requires human escalation.

Module 3: Reasoning under Uncertainty

You have to develop a probabilistic model for estimating the likelihood of fraud and the reliability of the available evidence. The model may consider consistency among customer statements, authenticity of submitted documents, transaction history, similarity to previously detected fraudulent cases, missing information, contradictory timestamps, and the reliability of external verification sources. Bayesian reasoning or another suitable uncertainty-handling technique should be used to update the probability of eligibility or fraud as new evidence becomes available. The system should distinguish between missing evidence, weak evidence, contradictory evidence, and evidence that appears deliberately fabricated. It should also avoid treating an unusual request as fraudulent merely because it differs from common cases.

Module 4: Reinforcement Learning

You have to formulate claim-handling decisions as a Reinforcement Learning problem. At each stage, the agent may approve the request, reject it, ask for additional evidence, perform another verification, propose an alternative resolution, or escalate the case to a human authority. The state should capture the applicable policy, satisfied and unsatisfied constraints, evidence reliability, estimated fraud probability, interaction history, and cost of further investigation. The reward function may consider correct resolution, financial loss, customer satisfaction, investigation cost, policy violations,

unnecessary rejection of genuine claims, and inappropriate escalation. You may implement Q-learning, SARSA, a contextual bandit, or another suitable reinforcement learning algorithm and compare it with fixed strategies such as always escalating uncertain cases or approving requests whenever the customer provides the required documents. Natural Language Understanding techniques should be used to extract claims, dates, amounts, supporting evidence, emotional appeals, threats, contradictions, and persuasive strategies from the conversation.

Module 5: Generative AI

You have to develop a multi-agent GenAI environment containing customer, customer-service, policy-verification, fraud-analysis, human-manager, and red-team agents. Cooperative customer agents should represent genuine users with valid or partially documented requests, while red-team agents should attempt to influence the service agent through emotional pressure, fabricated evidence, inconsistent narratives, or misleading interpretations of policy. The customer-service agent should communicate naturally and empathetically while relying on the search, CSP, planning, uncertainty, and reinforcement learning modules for its decisions. The policy-verification agent should identify the applicable rules, the fraud-analysis agent should examine suspicious evidence, and the manager agent should review cases requiring human approval. The final response must clearly explain the decision, the policy clauses considered, the evidence that was accepted or questioned, and any additional action available to the customer.

Expected Outcome

The final system should demonstrate how a customer-service agent can remain helpful and respectful while resisting attempts to manipulate it into violating organisational policies. You should evaluate the system using both genuine and adversarial scenarios, including incomplete documentation, legitimate exceptional circumstances, emotional appeals, conflicting statements, fabricated evidence, ambiguous policy clauses, repeated claims, and attempts to obtain unauthorised refunds. A successful implementation should correctly resolve eligible requests, reject or investigate suspicious requests, escalate genuinely uncertain cases, and provide transparent policy-grounded explanations. The evaluation should consider policy-compliance rate, fraud-detection performance, false rejection of genuine users, financial cost, number of verification actions, escalation frequency, and robustness against different persuasive strategies.

References

Fang, Haishuo, Xiaodan Zhu, and Iryna Gurevych (2025). “Preemptive detection and correction of misaligned actions in llm agents”. In: *Proceedings of the 2025 conference on empirical methods in natural language processing*, pp. 222–244.

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